Chapter 8. Main memory. Exercises

Original problems. State assumptions, units, and intermediate reasoning. Answers are in the separate solutions PDF.

OS08-E01. Page translation

Page size is 4 KiB. Virtual address 0x2A7C has page number 2. The entry maps page 2 to resident frame 9 with read-only permission.

1. Compute offset and physical address.
2. What happens on a write?
3. If the entry is not currently usable, must that imply a design error?

Check your approach. Retain offset width and permission checks throughout the arithmetic.

OS08-E02. Effective access time

A serialized TLB lookup costs 2 ns and a memory access 100 ns. A TLB miss requires one additional page-table memory access. Ignore page faults. Let h=0.95.

1. Derive EMAT and evaluate it.
2. Recompute at h=0.80.
3. How should page-fault cost enter the model?

Check your approach. Enumerate the actual hit and miss operations before weighting them.

OS08-E03. Page-size tradeoffs

There are 100 objects of 33 KiB each. Each object has its own page-aligned allocation and cannot share its tail page with another object. Compare 4 KiB pages with a 64 KiB alternative.

1. How much tail space does one 4 KiB-page allocation waste?
2. Why can 64 KiB pages reduce entry count?
3. Are smaller pages always more memory-efficient overall?

Check your approach. Include both space and translation costs, with their allocation assumptions.

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Chapter 8. Main memory. Exercises (continued)

OS08-E04 An unfamiliar x86-64 walk

Four levels,4 KiB pages,48-bit canonical VA. CR3=0x7000; PDPT/PD/PT bases 0x8000/0x9000/0xA000. VA 0x00001234_5678_9ABC, frame 0x12AB. Every level present/user/RW, NX0.

1. Check canonical and give all four indices and offset.
2. Use 8-byte entries to find entry addresses and plausible values.
3. Derive PA.
4. Keep the same mapping but set PTE 0x12AB005. Give read/write results.

OS08-E05 Canonical, present, and permissions

Use E04's four-level format. Test (A) VA 0x00002468_ACE0_1234,(B) a canonical path with PML4E.P=0,(C) the exact same mapping with PTE.NX=1 on fetch,(D) VA 0x00008000_00000000.

1. Give A's indices/offset and canonical result.
2. Which case rejects before walking and which faults at a table level?
3. Which data reads/writes succeed and which fetches fault?
4. Can a present-only check distinguish write protection/NX? Give a complete operation table or executable branch model.

OS08-E06 ARM layout, domain, and AP

ARMv7-A AArch32 short descriptors, N0, AFE0, TRE0,4 KiB small pages. P1 root 0x14000, L2 0x1C000; P2 root 0x18000, L2 0x1C400. L1 raw 0x1C061/0x1C461, L2 both 0x54862, D3 Client. P1 VA 0x004A6000, P2 VA 0x007C8000, refs 2.

1. Prove storage sizes/alignment/nonoverlap before finding L1/L2 indices and four entry addresses.
2. Decode all raw fields, using bit 9 and bits 5:4 for small-page AP. Give both processes' read/write results.
3. Remove P1 mapping. Give P2 read, P1 read, refcount results.
4. Change D3 to Manager or AFE to 1. Does the original write-fault conclusion still hold?

Chapter 8. Main memory. Exercises (continued)

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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